

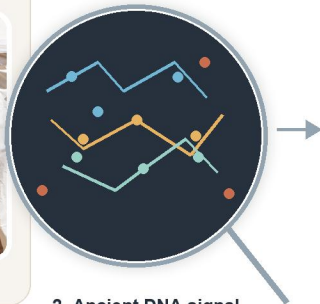
Ancient DNA source-data audit scenario

From archaeological material to auditable local QC, provenance checks, and expert interpretation.

1. Archaeological context



Physical context, accession metadata, and sample provenance define the boundary of downstream analysis.



2. Ancient DNA signal

fragmentation · damage · contamination risk

3. Agent-assisted audit

Source records

ENA / SRA metadata

FASTQ QC

read count · GC · length

Expert review

provenance and limits

Audit outputs

reports and records

The agent checks whether local files match public records before biological interpretation.

Claim boundary: source-data reproducibility,
not biological proof.

Graphical message: ancient material produces fragile molecular data; the agent turns source files into auditable checks while experts retain interpretation.